

Expanding Recurrence Relation

Definition:

Expanding a recurrence means repeatedly replacing the recurrence relation with itself to reveal its structure.

Expand step by step

- * Write the recurrence
- * Replace the term using the same recurrence
- * Continue expanding until a pattern appears

Eg:

$$T(n) = T(n-1) + 2$$

$$T(n) = T(n-2) + 2 + 2$$

Identify Pattern

- * Observe repeated terms & their coefficient.
- * Look for arithmetic, geometric or other consistent patterns.
- * Generalize the pattern to guess a formula.

Eg:

$$T(n-k) = T(n-k) + \underbrace{(1+1+\dots+1)}_{k \text{ times}}$$

Simplify Expression

- * Use the identified pattern to write the sum in closed form
- * Combine terms & simplify the expression
- * Express the recurrence in terms of base case(s).

Eg: $T(n) = T(0) + cn$

(for $T(n) = T(n-1) + c$)

Substitution Method for solving.

1. Assume a solution form



Assume a general form based on the pattern
Eg: $T(n) = an + b$

Substitute into Recurrence



Substitute the assumed form into the original recurrence.

Solve for Constants



Equate coefficients / compare terms to find the values of unknown constants

Verify by induction



Use mathematical induction to verify the solution for the base case and inductive step.

Importance of Recognition Pattern

- * Reduce repeated calculations & save times.
- * Transforms complex recurrences into simple forms
- * Helps derive closed form solutions
- * Enhances understanding of the behaviour of $T(n)$
- * Improves accuracy & efficiency in proofs & analysis

How this map Aids in solving effectively

- * Provides a clear, step by step approach.
- * Connects expansion, pattern recognition, substitution & verification seamlessly
- * Helps in guessing the right form & verifying it correctly.
- * Works for a wide range of linear & non linear

Final outcome:



Accurate, simplified & general (closed form) solution of recurrence relations.
Enables effective analysis of algorithms & mathematical models.